# Client Requirements

Kalahari Biodiversity Research Centre | Project ID: CMPG325-2026-107 | Client ID: CLI-107

## 1. Client Overview

The Kalahari Biodiversity Research Centre, based in Upington, is a research organisation focused on studying and monitoring biodiversity within the Kalahari region. Its operations involve a combination of administrative functions (staff management, finance, and month-end reporting) and specialised research activities (data collection, analysis, and coordination among field and laboratory staff). Given this dual nature, the network must reliably support both routine office operations and data-sensitive research work, while remaining stable and secure enough to protect ongoing research and administrative processes.

## 2. Functional Requirements

Based on the client brief, the network must provide the following:

- Segmented connectivity for distinct organisational functions — Administration, Research/Laboratory operations, centralised Servers, and Voice communications — ensuring each group has appropriate access without unnecessary exposure to unrelated traffic.
- Internal DNS (Domain Name System) service, allowing staff to access internal network resources (such as file or data servers) using human-readable hostnames rather than memorised IP addresses. This improves usability and reduces configuration errors when internal services are referenced or updated.
- IP addressing based on the assigned block 192.168.48.0/24, structured to support all departments in a logical, scalable, and well-documented manner.
- Support for VoIP handsets (per Change Request CR11), requiring the network to accommodate voice traffic alongside existing data traffic without degrading either service.
- A working, testable Cisco Packet Tracer implementation demonstrating all of the above in a simulated environment.

## 3. Non-Functional Requirements

The client has specified a critical operational constraint: no network downtime is tolerated during month-end processing. This is understood to primarily affect the Administration function, which handles finance and reporting activities at month-end. To address this, the network design isolates Administration traffic within its own VLAN and subnet, reducing the likelihood that issues originating in other segments (such as Research or Voice) could impact Admin connectivity during this critical period. This design choice reflects a broader principle of fault containment through network segmentation, rather than relying on a single, undifferentiated network where a fault anywhere could affect operations everywhere.

## 4. Scope Statement

This design addresses the specific requirements of Client CLI-107 (Kalahari Biodiversity Research Centre), Project ID CMPG325-2026-107, as outlined in the approved project brief. It reflects the assigned addressing block (192.168.48.0/24), the assigned technical challenge (DNS — internal name resolution), and Change Request CR11 (VoIP accommodation). No alternative client scenario has been substituted in this solution.